How Does Technology Affect our Relationship with God?

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1 Introduction

Within the field of Human-Computer Interaction (HCI), significant research has focused on designing systems, platforms, and content that capture and sustain user attention. Modern Smartphones employ persuasive design strategies, personalized algorithms, and continuous notifaction systems to maximize engagement. While these technologies have increased accessibility, communication, and productivity, they have also contributed to unprecedented levels of screen time and digital dependence.

As engagement with digital devices become more constant, researchers have begun examining the broader implications of sustained technological interaction. Much of the existing literature explores impact on mental health, productivity, and interpersonal relationships. However, comparatively less attention has been given to how these technologies influence spiritual life and religious practices, specifically in Christianity. From both psychological and theological perspectives, humans are often described as relational beings whose well being depends on intentional connection with others (e.g., Genesis 2:18), and for many, with God. Practices such as prayer, solitude, worship, and reflection require sustained attention and cognitive presence, qualities that are challenged by distracting environments.

This paper investigates the question: How does technology affect our relationship with God? This will be done by a literature review that examines research in HCI, Customer Smartphone Distraction(CSD), Technology in spiritual disciplines, Prayer, and Scripture reading. By synthesizing these findings, this review identifies emerging themes, points of agreement, and areas where further research can be done.

2 Background

To better understand the relationship between technology and spiritual life, it is important to examine key concepts that shape this discussion. The following section provides background on Human Computer Interaction, smartphone distraction, and Christian spiritual disciplines, establishing the conceptual foundation needed to evaluate exisiting research in these areas.

2.1 HCI

Human Computer Interaction is an interdisciplinary field concerned with the design, evaluation, and implementation of interactive computing systems for human use. A central area of inquiry within HCI examines how interface design influences user behavior and attention[2]. Design elements ranging from visual layout and button placement to notification systems and algorithmic content delivery shape how users engage with digital environments. While HCI research has traditionally emphasized usability, efficiency, and health applications, more recent work has begun exploring religion and spirituality as potential design spaces. However, these areas remain comparatively understudied within mainstream HCI scholarship.

When it comes to HCI, being human-centered is extremely important, in fact it's considered to be one of the pillars of HCI as a discipline [3]. But what does it mean to be human-centered? Human-centered design means that the choices being made in a design process are being made to tailor to the user's needs, goals, and preferences. This is done by understanding the user through research and then using that understanding to inform design decisions.

While it's important to be human-centered in design, in order to be human-centered, you have to understand how to do user research. User Research is the process of understanding the methods used to obtain actionable knowledge about users, including their activities, needs, wants, and contexts of use [3]. User Research stems from the realization that the designer and the user are not the same person and that the designer cannot assume to know what the user needs or wants without first understanding them. User research can be done in a variety of ways, including interviews, surveys, observations, and usability testing.

Another important concept in HCI is interaction. Interactions can describe many different things, like individuals, groups, and communities using computers. Interactions also take place in different timescales, like pressing a button, or onboarding employees to a new software system. To put it more generally, interaction is a dynamic phenomenon that unfolds over time as users and computers influence each other.

User Interfaces are the means through which users interact with computers. One source describes user interfaces as "the only part of a system that is used by real people doing real tasks in real settings. Therefore, what makes an interface good is ultimately determined by humans. As a result research on user interfaces is a cross-disciplinary effort with contributions from computer science, electrical engineering, design, and psychology" [3]. User interfaces can be designed in a variety of ways, but the goal is to create an interface that is efficient, and effective. Another goal for most interfaces is to make it easy to learn and use, but take a website like Bloomberg Launchpad for example, which is a financial news website. It is not designed to be easy to learn and use, but rather to be efficient and effective for its target audience, which is financial professionals. This is an example of how user interfaces can be designed with different goals in mind, and that the design of an interface can have a significant impact on how users interact with it.



Figure 1: Bloomberg Launchpad Interface

Design is also a central concept in HCI. Design can be described as the process of making choices about high level concepts or minute details to enhance the user experience. Design affects the way we perceive, experience, remember, decide, and form habits. Design can make or break someone's ability to use something, even small details can have significant impacts.

2.2 CSD

One important concept relevant to this discussion is Customer Smartphone Distraction, which is defined as an internet-connected smartphone competing for and compromising a user's attentional resources that were originally directed toward another task[7]. Mobile Devices are frequently engineered to maximize engagement through persistent connectivity, notifications, and personalized content feeds. As a result, they may fragment attention and interrupt sustained cognitive or reflective activities. Given that many spiritual practices require focused attention and intentional presence, this dynamic raises important interdisciplinary questions.

2.3 Spiritual Disciplines

In this paper, spiritual practices are discussed within a Christian framework. Spiritual discipline refers to structured activites intended to cultivate a relationship with God, including prayer, Scripture reading, worship, solitude, silence, and rest. Prayer has been defined as "every kind of inward communion or conversation with the divine power"[1]. A quote from Finney and Malony that defines prayer in a Christian context is, "In Christianity God is seen to be essentially all-loving and all-powerful, the one who has revealed himself in Jesus, The Christ. In the Christian tradition prayer is, this, intimate communication with a benevelont deity." This study of prayer suggests that such practices involve focused cognitive and emotional investment, further underscoring the importance of sustained attention in spiritual life.

Worship functions as a broader category encompassing various forms of devotional expression, including singing, praise, thanksgiving, and service. While these practices vary in format, they share an emphasis on intentional orientation toward the divine. As such, they provide a meaningful lens through which to examine how technological environments, particularly those optimized for interruption and engagement, may influence patterns of spiritual practice and relational presence.

2.4 Technology in Spiritual Disciplines

In addition to shaping patterns of attention, digital technologies also function as platforms through which spiritual practices may be facilitated. Mobile applications provide access to Scripture, guided prayer resources, devotional content, and livestreamed worship services, extending religious participation beyond traditional physical spaces[8]. For individuals with limited access to in-person communities, digital environments may enable forms of connection and engagement that would otherwise be unavailable. From an HCI perspective, this reflects the capacity of interactive systems not only to capture attention, but also to mediate meaningful experiences, including those related to faith and religious practice. As a result, technology's role in spiritual life is not solely disruptive, but complex and multifaceted.

Together, these perspectives suggest that technologies designed to capture and sustain attention may simultaneously enable and complicate practices that depend upon sustained relational presence. This tension forms the foundation for examining how technology may affect an individual's relationship with God.

3 Related Works

Having established key concepts surrounding engagement driven design, attention fragmentation, and Christian spiritual disciplines, the next section examines exisiting research that explores the intersection of these areas.

3.1 HCI

Research in HCI increasingly examines how interactive systems are intentionally designed to optimize user engagement and sustain attention. Studies within HCI highlight how interface elements such as notification systems, personalized content delivery, and feedback loops reinforce habitual interaction patterns and extend user engagement [2, 8]. These design strategies are frequently framed as enhancing usability, responsiveness, and user satisfaction by reducing friction and encouraging continued participation. Further work in persuasive and reflective design suggests that digital environments not only respond to user behavior but actively shape attentional habits through structured interaction flows and engagement incentives [4]. Together, this body of research positions attention optimization as a central principle in contemporary digital systems, establishing engagement as both a design objective and a behavioral outcome of interface architecture.

Building on this, human centered design emphasizes multiple dimensions of understanding people, including perception, motor control, cognition, needs, experience, communica-

tion, and collaboration. [3]. Perception concerns how users interpret information through their senses, which directly informs the visual and sensory design of user interfaces. Motor control focuses on how individuals plan and execute physical movements, influencing the design of input devices and interaction techniques. Cognition encompasses processes such as memory, attention, and reasoning, and plays a critical role in the design of complex interactive tasks that require users to process and retain information. In addition to these cognitive and physical factors, human centered design also considers users' needs, which reflect the goals and motivations that drive behavior. Understaning these needs allows designers to prioritize what is meaningful and relevant to users in both specific tasks and broader contexts. Experience refers to how users perceive and interpret their interactions over time, highlighting the importance of designing systems that create positive and meaningful experiences. Communication examines how people exchange information with one another and with systems, informing the design of applications that support human to human and human computer interaction. Finally, collaboration focuses on how individuals work together, guiding the development of systems that facilitate coordination, cooperation, and shared tasks.

Expanding on the concept of user research, it is important to understand the different methods used to obtain actionable knowledge about users. These methods include interviews, surveys, observations, and usability testing. Interviews allow researchers to learn about the subjective experiences of users. The purpose is to learn how they view their tasks, values, experiences, and practices, and what their wishes are for interactive systems[3]. Observation is a method of data collection where a researcher observes and takes notes on activities related to the use of an interactive system. This method will be used in this study to observe how users interact with the application and to gather data on their behavior and engagement patterns. Surveys will also be used to gather qualitative and quantitative data on users' perceptions of attention, distraction, and reflective engagement during their interactions with the application. In Survey Research, the researcher designs a questionnaire and distributes it to respondents. A questionnaire is a series of questions presented and answered in a structured way. Surveys can be used to understand users' behaviors, routines, or use of interactive systems. They can also be used to understand user experiences, needs, wants, and desires, as well as attitudes, preferences, and beliefs [3]. Usability testing is a method of evaluating a product or system by testing it with representative users. The goal of usability testing is to identify any usability problems, collect qualitative and quantitative data, and determine the participant's satisfaction with the product. This method will be used in this study to evaluate the effectiveness of the application in reducing distraction and promoting attentiveness during spiritual practices.

Diving deeper into the concept of interaction, it is important to cover some theories of interaction. The first is Information and Control. This is an interaction that is viewed as the user and the computer send messages across a noisy channel to communicate and control each other. This interaction happens within seconds. The second theory is Dialogue. This is structured as communication turns between the computer and the human, with each turn changing the context of the subsequent turn. This interaction happens in seconds to minutes [3]. The third theory is Tool use. This involves users using computers as tools to accomplish tasks. This can take minutes to days. The fourth theory is Automation. These interactions involve users delegating tasks to Artificial Intelligence systems that can

perform tasks on their behalf in order to complete goals quicker. These interactions can take minutes to days. The fifth theory is Rationality. Rationality is about users choosing actions with a computer that they believe will maximize their utility. Success can be limited by the computer and the users' abilities. The time span for these interactions can take anything from seconds to weeks. The sixth theory is Practice. Interactions are situated in and affected by organizational and societal practices [3]. This can take anywhere from weeks to years. These theories of interaction provide a framework for understanding the different ways users interact with computers and how these interactions can be designed to optimize user experience and engagement.

Looking more into user interfaces, they can be broken down into four elements: Devices, Interaction techniques, Representations, and Assemblies. Devices, or Peripherals, are the physical components that allow users to interact with computers like a mouse, keyboard, or microphone. Interaction techniques translate input from peripherals into actions that the computer can understand, and the user can see the results of their interactions. The gain function, which maps mouse movement to cursor movement, is an example of an interaction technique [3]. Representations determine how data, events, objects, and actions appear to the user. Representations can be visual, auditory, or haptic, which means they can be seen, heard, or felt. A visual icon like the trash can would be an example of a representation. Finally, Assemblies describe the broader set of principles that organize represenations. Assemblies are quite intuitive, they define the way that represenations are assembled in a user interface so that they are easy to understand and use. An example of this would be the way pages may be linked on a website, or the way that a menu is organized on an application.

Delving into design, it can be broken down into many different categories, but for this paper we will focus on Interaction and user centered design. Interaction design can be summarized into four rationales: Improving, Creating, Informing, and Producing knowledge [3]. Improving is about design being able to improve the quality of life for users when experiencing computing systems. Creating is about how design can create entirely new opportunities for people to do things with computers that they could not do before. Informing is the idea that design is the study of what is possible. By showing new possibilities, design informs decisions and creates the need for change. Producing knowledge is about how design not only applies HCI knowledge but is necessary for producing new knowledge in HCI [3]. Knowledge in interaction can be improved by testing ideas through usability testing, surveys, and observations. User Centered Design can be summarized into User focus, User involvement, Iterative and incremental development, Prototyping, Evaluation in context, Holistic design, and Process customization [3]. User focus is about how the design process should be focused on serving the user's goals, tasks, and needs. This concept alone distinguishes user centered design from many other processes in engineering and business. User involvement is about how users should be involved in the design process, and that their feedback should be used to inform design decisions. Iterative and incremental development is about how it's impossible to get a design right the first time, and that the design process should be iterative, with multiple rounds of design, testing, and refinement. Prototyping helps evaluate ideas and concepts in the design process before converging to a particular solution. Evaluation in context is about how design should be evaluated by real users in real contexts, rather than in a lab setting. This is important because it allows designers to understand how users interact with the design in their natural environment, which can reveal insights that may not

be apparent in a controlled setting. Holistic design is about understanding that all designed aspects of a product affect its use and user experience, like manuals, social media presence, and advertisements [3]. Finally, process customization stresses the importance of avoiding a one size fits all approach to design, and instead ensuring that there is continuous reflection and evolution of the design process itself to fit the needs of the project and the users.

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3.2 CSD and Cognitive Fragmentation

Empirical studies on smartphone distraction and cognitive fragmentation demonstrate that attentional disruption emerges from an interaction between device design features, psychological states, and environmental stimuli[7]. Research indicates that notifications, app design, and constant connectivity contribute to fragmented attention, reducing users' ability to sustain focus on tasks or prolonged cognitive activities. Similarly, studies examining technology's effect on Christian practice suggest that pervasive smartphone usage distract and take away from core religious values and reshape patterns of engagement with spiritual teachings. It has also been argued that habitual reliance on digital platforms can influence behavior and priorities, potentially altering the expression of Christian identity[6]. Together, these studies suggest that while smartphone design effectively captures and sustains user attention, this engagement may unintentionally disrupt activities requiring pronlonged focus, including spiritual practices. This tension between attention optimizing technologies and religious practices that depend on sustained focus underscores the need to examine how device mediated environments interact with religious and contemplative behaviors.

3.3 Spiritual Disciplines

Christian spiritual practices are traditionally understood as intentional activities that cultivate reflection, attentiveness, and relational engagment with God. Common disciplines include prayer, scripture reading, worship, silence, and solitude, all of which require sustained focus and intentional participation. Prayer, for example, has been described as a form of inward communion or conversation with the divine, emphasizing a relational interaction between the believer and God [1]. Within Christian theology, this interaction is often framed as intimate communication with a benevolent and personal deity, making attentiveness an essential component of the practice. Similarly, engagement with scripture involves interpretive reflection and critical understanding rather than passive consumption of information. Research on Bible engagement highlights the importance of thoughtful reading and hermeneutical interpretation in fostering spiritual understanding and personal transformation [5]. Together, these practices emphasize sustained attention and intentional reflection as central components of Christian spiritual life.

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3.4 Technology in Spiritual Disciplines

While concerns about distraction are significant, research also suggests that digital technologies can play a constructive role in supporting spiritual practices. Recent work in HCI has begun to explore religion and spirituality as potential spaces for thoughtful design, examining how digital tools can facilitate reflection, prayer, and engagement with religious communities [2]. Mobile applications, online worship services, and digital scripture platforms allow individuals to access religious resources more readily and maintain connection with faith communities across geographic boundaries. In some cases, these technologies provide new opportunities for spiritual engagement by integrating reminders for prayer, guided devotional content, or community discussion into everyday digital environments. Rather than functioning solely as sources of distraction, these systems demonstrate that technology can also mediate and support spiritual practices when designed with reflective or contemplative goals in mind. As a result, the relationship between digital technology and spirituality is not purely oppositional but represents a complex interaction between engagement driven design and the intentional practices that characterize religious life.

3.5 Conclusion of Sources

Taken together, the literature reviewed above highlights a potential tension between the engagement driven nature of modern digital technologies and the attentional demands of Christian spiritual practices. Research in Human Computer Interaction demonstrates how contemporary digital systems are designed to maximize user engagement and sustain interaction through persuasive interface design. At the same time, studies of smartphone distraction suggest that these systems can fragment attention and disrupt sustained cognitive focus. In contrast, many Christian spiritual disciplines, such as prayer, scripture reading, and reflective worship, emphasize intentional attentiveness and prolonged reflection as central components of spiritual life. Although some technologies have been developed to support spiritual engagement, the broader design of digital environments often prioritizes continuous interaction rather than contemplative focus. As a result, the intersection of digital technology and Christian spiritual practice presents an important area for further exploration, particularly regarding how engagement oriented technologies may shape or influence individuals' ability to participate in spiritually reflective activities.

4 Methodology

4.1 Research Questions

This study is guided by the primary research question: How does technology affect an individual's relationship with God? To explore this question, a secondary research question is also considered: Can software application be designed to support spiritual disciplines and mitigate the distracting effects of technology on religious practice? These questions aim to investigate both the potential negative impacts of technology on spiritual life and the possibility of leveraging digital tools to enhance religious engagement. By addressing these questions, this study seeks to contribute to a deeper understanding of the complex

relationship between technology and spirituality, particularly within the context of Christian practice.

To investigate these questions, this project adopts a design based approach through the development of a software prototype. Rather than solely analyzing existing technologies, this approach allows for the exploration of how specific design choices, particularly those that limit engagement and reduce distraction, may influence user behavior and patterns of interaction. The software developed in this study is intended to function as a simplified, minimalistic interface that encourages reduced screen time and more intentional device use. By examining how such a system may alter engagement patterns, this project aims to provide insight into how technological design can shape attention and, by extension, practices that depend on sustained focus.

4.2 Hypothesis

Due to the short-term nature of the user study, these hypotheses focus on users' immediate perceptions of attention, distraction, and reflective engagement rather than long-term behavioral outcomes.

Based on the literature reviewed in HCI, smartphone distraction, and the attentional demands of Christian spiritual practices, the following hypotheses are proposed for evaluation during a controlled user testing session:

- H1: Users interacting with a minimalistic interface designed to limit engagement will report lower levels of perceived distraction compared to users interacting with a standard, engagement driven interface.
- H2: The absence of typical engagement features (e.g., notifications, personalized content) will lead to increased self-reported attentiveness and intentionality during the use of the application, as measured by post-interaction surveys.
- H3: Users will express a greater sense of reflective engagement and spiritual presence when using the minimalistic application, as compared to their typical interactions with standard digital platforms, as measured through survey responses and qualitative feedback during the user study.

4.3 Study Design

Participants in this study consisted of undergraduate students recruited from a christian club. No prior experience with the application was required. All participants were assumed to have general familiarity with smartphone usage and mobile applications. Participation was voluntary, and responses were collected anonymously.

This study employed a task based usability evaluation to investiagate how users interact with a minimalistic mobile application designed to support focused and reflective activity. Participants were asked to complete a series of structured tasks within the application God Time, simulating typical usage scenarios. Following task completion, participants completed a post task survey designed to capture their perceptions of usability, distraction, attention, and reflective experience. The study was conducted in a single session format, allowing for

the evaluation of immediate user experience rather than long term behavioral change. This approach aligns with the study's focus on perceived attention and interaction during short term use.

Participants completed a sequence of tasks designed to evaluate different aspects of the application's functionality and interaction design. First, participants were asked to explore the app freely for a short period in order to develop an initial understanding of its purpose and primary features. Next, participants initiated a standard stillness session using default settings and allowed the session to run to a completion. During or following this session, participants completed a reflective activity within the application by writing and submitting a short reflection. Finally, participants engaged with a "Strict Mode" stillness session. During this session, they were instructed to briefly leave the application and then return, allowing observation of how the system responded to interruptions in user engagement. These tasks were designed to capture both natural interaction patterns and responses to specific system features, particularly those related to attention and distraction.

Data was collected through the post task survey which was administered using the Qualtrics platform. They survey consisted of a combination of Likert scale and open ended questions designed to evaluate participants' experiences with the application. The Likert scale questions used a five point scale ranging from "Strongly Disagree" to "Strongly Agree" and focused on key aspects of user experience. In addition, open ended questions were included to gather qualitative feedback regarding user perceptions, preferences, and comparisons to typical smartphone usage.

5 Implementation

5.1 Resources Used

The application developed for this study, God Time, was implemented using TypeScript for application logic and CSS for interface styling. TypeScript was selected to provide a structured and type-safe environment for managing application state and user interactions, improving code reliability and maintainability. CSS was utilized to create a clean, minimalistic interface that aligns with the design goals of reducing engagement and promoting intentional use.

The application was built using React Native, a cross-platform framework that allows for development across both IOS and Android devices and uses a component based architecture that promotes modularity and reusability. This choice was made for its ability to efficiently create interactive user interfaces while maintaining performance across different platforms. The development process involved iterative design and testing to ensure that the application effectively supports the intended user experience of reduced distraction and increased attentiveness during spiritual practices.

For testing and deployment during development, Expo Go was used as a supporting framework for React Native. Expo Go provides a streamlined development workflow, allowing for rapid prototyping and testing on physical devices without the need for complex native build configurations. This facilitated the iterative design process, enabling quick adjustments based on testing results.

Progress was tracked using GitHub, which provided version control and collaboration features. This allowed for organized management of code changes, issue tracking, and documentation throughout the development process. By leveraging these resources, the application was developed efficiently while adhering to the design principles aimed at fostering a more intentional and less distracting user experience.

5.2 Design Choices

A central design goal of God Time was to minimize user interaction and reduce engagement driven behaviors commonly found in modern mobile applications. Rather than encouraging prolonged use, the application was intentionally designed with limited functionality and a simplified interface to promote intentional use. This approach aligns with the study's focus on reducing distraction and supporting sustained attention during spiritual practices.

The primary feature of the application is a stillness timer, which allows users to select a set duration for engaging in spiritual disciplines such as prayer, reflection, or scripture reading. Once initiated, the timer functions without additional interaction, encouraging users to disengage from the device and focus on the intended activity. This design choice contrasts with typical engagement oriented applications that continuously prompt user interaction through notifications, personalized content, and dynamic interfaces. The timer has two modes. A standard mode that allows users to leave the app and use their phone as normal, and a strict mode that tracks if the user leaves the app and ends the timer if they do. This design choice was made to allow users to choose the level of engagement they are comfortable with while still providing an option for those who want a more restrictive experience.

In addition to the timer functionality, the application includes a reflection journal feature that a user is guided to directly after their stillness session. This feature selects a question at random from a small pool to prompt users to reflect on their experience and spiritual engagement. The journal is designed to encourage thoughtful reflection without requiring additional interaction during the stillness session itself. By separating the timer and reflection features, the application supports a focused period of spiritual practice followed by intentional reflection time to foster deeper engagement with the experience overall.

6 Results

Findings showed that...

7 Conclusion

References

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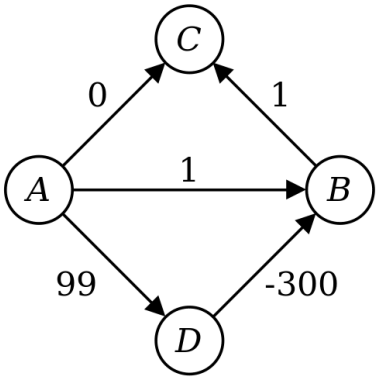


Figure 2: Dijkstra's algorithm does not work on graphs with negative edge weights, as evidenced here.